

International Cybersecurity and Digital Forensics Academy (ICDFA)

SBT-DF202: COMPUTER AND DIGITAL FORENSICS

Lab 2: USB IMAGE ACQUISITION AND HASH VERIFICATION

Student Practical Laboratory Manual

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EXECUTIVE SUMMARY

This practical laboratory exercise focused on the forensic acquisition and integrity verification of a USB flash drive using Exterro FTK Imager 8.3.0.27 on a Windows 11 Pro workstation.

A Kingston DataTraveler 3.0 USB device was used as the controlled test evidence. Before acquisition, an evidence folder named CIP-B102-Lab4-Evidence was created on the USB drive. A text file named kafayat_animashawun.txt was created within the folder containing the student's name, registration number, date, and a statement confirming that the device was prepared for a controlled forensic acquisition laboratory exercise.

The USB was identified as \\.\PHYSICALDRIVE1, with a reported source data size of approximately 59,006 MB. FTK Imager was used to acquire the entire physical device in Raw/DD format. The acquired image was stored on the local D: drive in D:\ICDFA\CIP-B102-Lab4.

The acquisition completed successfully, and FTK Imager subsequently verified the image. The calculated MD5 hash was 2c6e2f1a60c49b49887dc53d2fdf092e, while the SHA-1 hash was 8a243d04ea4e27fe5d40ac5156500262e6bd8d74. Both hashes were reported as verified matches. FTK Imager also reported that no bad blocks were found in the image.

The acquired image was subsequently loaded into FTK Imager, where the CIP-B102-Lab4-Evidence folder and kafayat_animashawun.txt file were successfully located. This confirmed that the expected evidence was preserved in the forensic image.

A SHA-256 hash was additionally calculated as required by the laboratory instructions and is documented in the hash verification section below.

PURPOSE OF THE ASSIGNMENT

The purpose of this practical was to demonstrate the fundamental principles and procedures involved in forensic USB disk imaging.

The exercise was designed to provide practical experience in identifying a digital storage device, preparing controlled evidence, creating a forensic image without modifying the original source, verifying the integrity of the acquired image using cryptographic hashes, and validating that expected evidence files were preserved within the acquired image.

The exercise also emphasized the importance of proper evidence documentation, repeatability, integrity preservation, and careful identification of the correct source device before performing low-level acquisition.

OBJECTIVES OF THE ASSIGNMENT

The objectives of this practical were to:

1. Explain the purpose and importance of forensic disk imaging.
2. Prepare a controlled USB test device for forensic acquisition.
3. Correctly identify the USB physical device before acquisition.
4. Document relevant evidence and acquisition information.
5. Acquire a forensic image of the complete USB device using an approved forensic tool.
6. Use Raw/DD format for the forensic acquisition.
7. Calculate cryptographic hashes for integrity verification.
8. Compare hash values associated with the source evidence and acquired image.
9. Verify that the acquired image was created successfully and contains no identified bad blocks.
10. Validate the acquired image by locating the expected evidence file.
11. Document the acquisition process using screenshots and evidence notes.

12. Identify risks associated with low-level forensic imaging tools and explain appropriate precautions.

METHODOLOGY OF THE ASSIGNMENT

The practical was conducted using a controlled test USB drive and the Windows version of Exterro FTK Imager. The methodology consisted of the following stages:

Stage 1: Evidence Preparation

A Kingston DataTraveler 3.0 USB flash drive was selected as the test evidence. A folder named CIP-B102-Lab4-Evidence was created on the USB drive. Within the folder, the file kafayat_animashawun.txt was created.

The file contained the student's name, registration number, date, and a statement confirming that the USB was prepared for a controlled forensic acquisition laboratory exercise.

Stage 2: Evidence Documentation and Identification

The physical USB was identified using FTK Imager. The device was identified as \\.\PHYSICALDRIVE1, with the model reported as Kingston DataTraveler 3.0 USB Device and a source data size of approximately 59,006 MB.

The identification process was documented before acquisition to reduce the risk of selecting the wrong storage device.

Stage 3: Forensic Image Acquisition

FTK Imager 8.3.0.27 was used to acquire the physical USB device. The source was selected as the identified physical drive, and Raw/DD was selected as the image format. The image was stored on the computer's D: drive rather than on the source USB:

D:\ICDFA\CIP-B102-Lab4

The image was created as a segmented Raw/DD image, with image segments generated sequentially from .001 through .040.

Image verification was enabled during the acquisition process.

Stage 4: Hash Verification

FTK Imager calculated MD5 and SHA-1 values during the acquisition and verification process. The resulting values were compared with the corresponding report values.

Additional SHA-256 hashing was performed to satisfy the laboratory requirement for SHA-256 integrity verification.

Stage 5: Image Validation

The acquired image was loaded into FTK Imager as an image evidence source. The CIP-B102-Lab4-Evidence folder was located within the acquired image, and kafayat_animashawun.txt was successfully identified.

The contents of the file were also viewed to confirm that the expected evidence had been preserved.

TOOLS AND RESOURCES USED

The following tools and resources were used during the practical:

Tool/Resource	Purpose
Exterro FTK Imager 8.3.0.27	USB identification, forensic acquisition, verification and image validation
Windows 11 Pro	Acquisition workstation operating system
Kingston DataTraveler 3.0 USB	Controlled test evidence
File Explorer	USB preparation and image storage management
PowerShell	Cryptographic hash calculation where applicable

5. EVIDENCE DOCUMENTATION

The evidence device was documented before acquisition.

Evidence Attribute	Recorded Information
Case/Lab ID	SBT-DF202-Lab2
Evidence Number	001
Student Name	Kafayat Omolara Animashawun
Student ID	2025/FWSD/11317
USB Manufacturer/Model	Kingston DataTraveler 3.0 USB Device

Interface	USB
Removable Drive	Yes
Source Type	Physical
Physical Drive	\\.\PHYSICALDRIVE1
Source Data Size	59,006 MB
Acquisition Tool	Exterro FTK Imager 8.3.0.27
Acquisition Format	Raw/DD
Image Storage Location	D:\ICDFA\CIP-B102-Lab4
Evidence File	kafayat_animashawun.txt
Evidence Folder	CIP-B102-Lab4-Evidence

USB Evidence Preparation

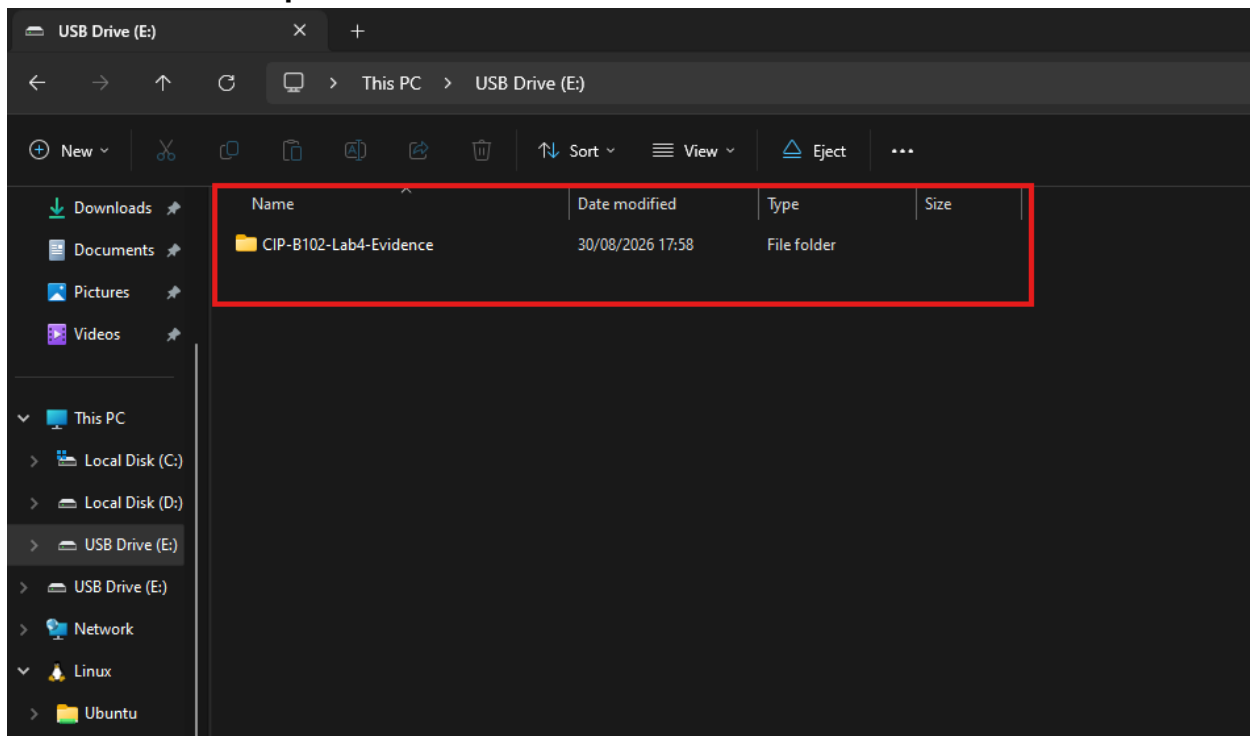


Figure 1: USB evidence folder containing the controlled laboratory evidence file.

6. CONTENTS OF THE STUDENT-CREATED EVIDENCE FILE

The file kafayat_animashawun.txt was created inside the CIP-B102-Lab4-Evidence folder.

The contents included:

- Full Name: Kafayat Omolara Animashawun
- Registration Number: 2025/FWSD/11317
- Date: 30 August 2026
- A statement confirming that the USB was prepared for a controlled forensic acquisition laboratory exercise.

Evidence File Content

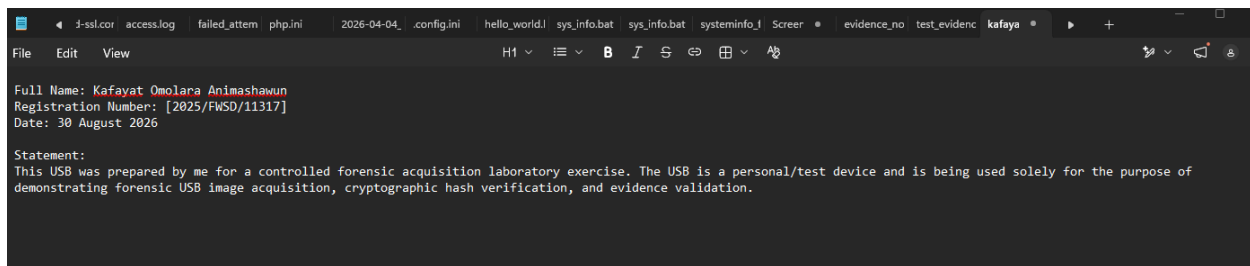


Figure 2: Contents of the student-created evidence file.

7. USB IDENTIFICATION

Before acquisition, the USB device was carefully identified in FTK Imager to ensure that the correct source device was selected.

The device was identified as:

Physical Device: \\.\PHYSICALDRIVE1

Model: Kingston DataTraveler 3.0 USB Device

Interface: USB

Removable: Yes

Source Data Size: 59,006 MB

The device identification was cross-checked against the physical USB being used for the laboratory exercise. The reported capacity and Kingston device information corresponded with the test USB.

Correct source identification was important because selecting an incorrect physical drive during forensic acquisition could result in accidental acquisition of another device or, when using write-capable low-level tools, possible data loss.

Correct USB Identification

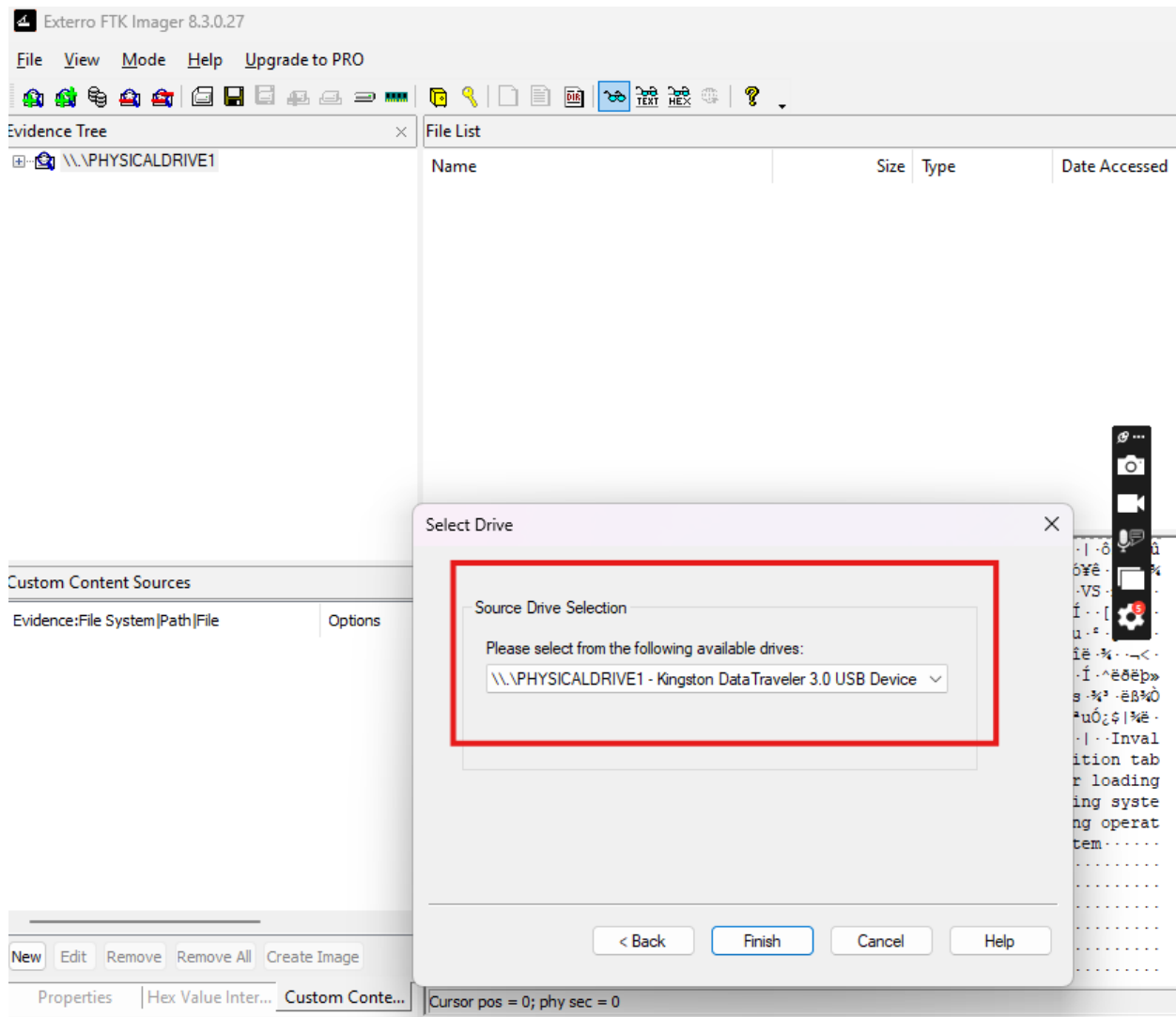


Figure 3: Identification of the Kingston DataTraveler USB as the source physical device.

8. FORENSIC IMAGE ACQUISITION

The forensic image was acquired using Exterro FTK Imager 8.3.0.27.

The source selected for acquisition was:

\\.\PHYSICALDRIVE1

The image format selected was:

Raw/DD

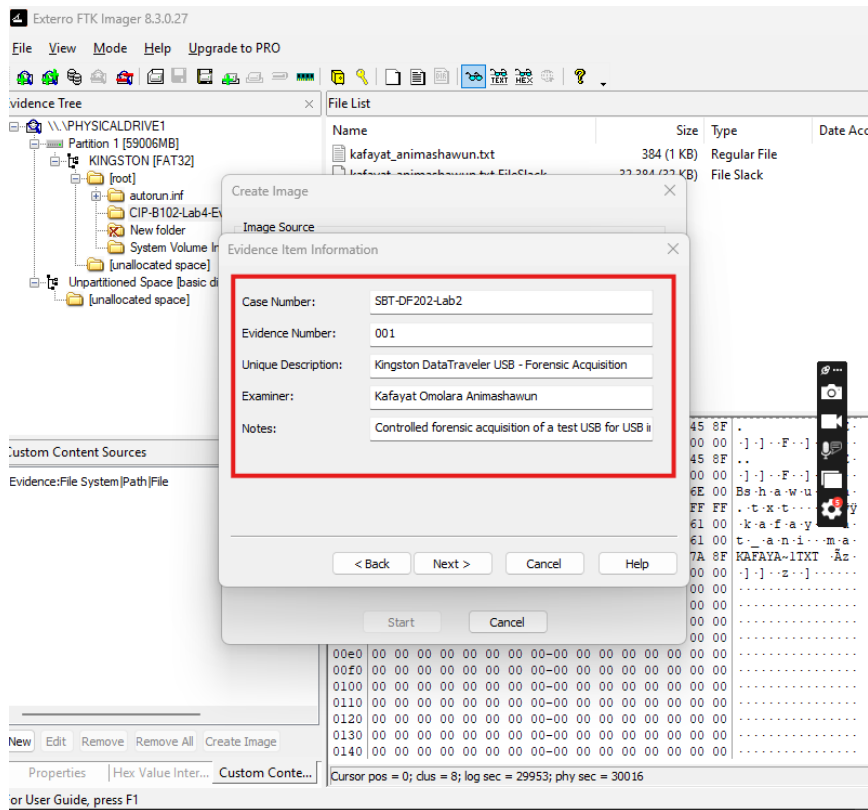
The evidence information entered into FTK Imager was:

Case Number: SBT-DF202-Lab2

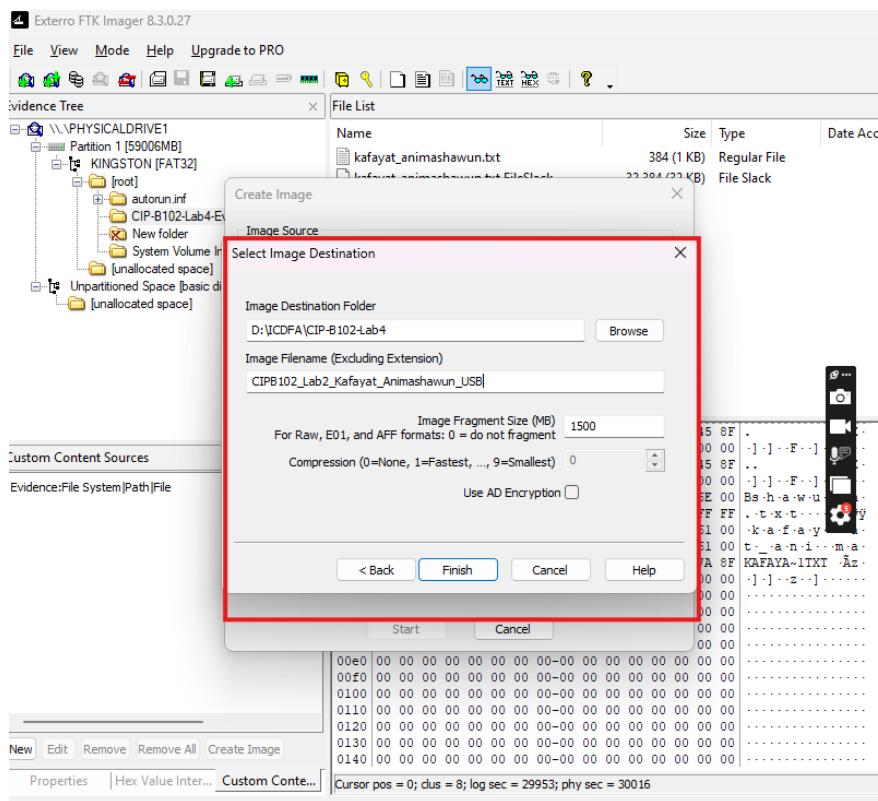
Evidence Number: 001

Unique Description: Kingston DataTraveler USB - Forensic Acquisition

Examiner: Kafayat Omolara Animashawun



The image was stored on the local computer at:
D:\ICDFA\CIP-B102-Lab4



The image destination was intentionally located on the computer rather than the source USB in order to avoid overwriting or modifying the source evidence.

FTK Imager generated the Raw/DD acquisition as multiple sequential image segments.

The generated image segments ranged from .001 through .040.

Image verification was enabled during the acquisition process.

FTK Acquisition Configuration

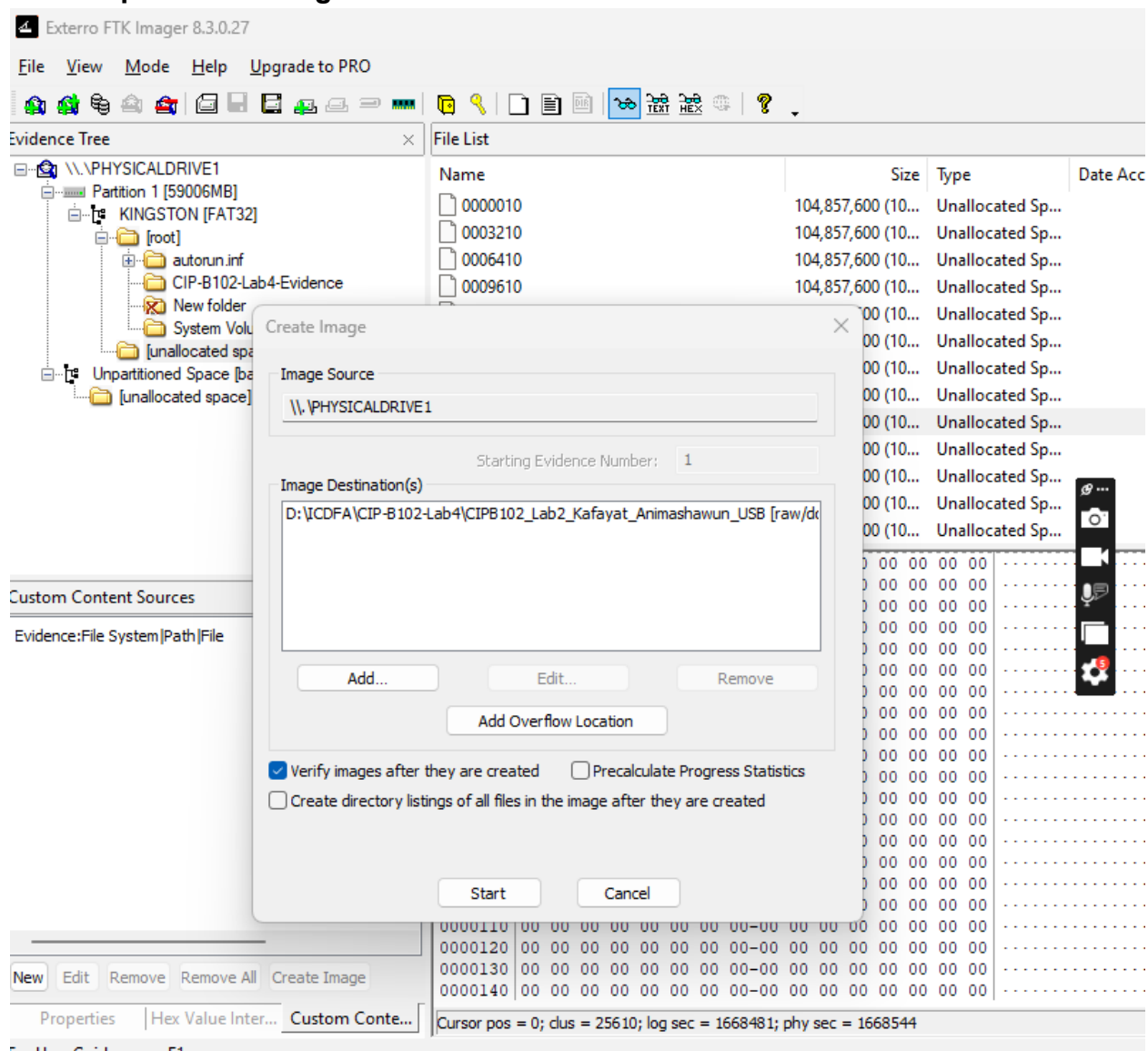


Figure 4: FTK Imager configuration showing the physical USB source, Raw/DD acquisition and image destination.

9. ACQUISITION COMPLETION AND VERIFICATION

The acquisition began at approximately 19:36:38 on 30 August 2026 and completed at approximately 19:45:46.

FTK Imager subsequently performed image verification from approximately 19:45:49 to 19:49:12.

The verification results reported that the calculated MD5 and SHA-1 values matched their corresponding report values.

FTK Imager also reported:

No bad blocks found in image.

This indicated that the acquisition completed successfully and that the generated image passed the verification performed by FTK Imager.

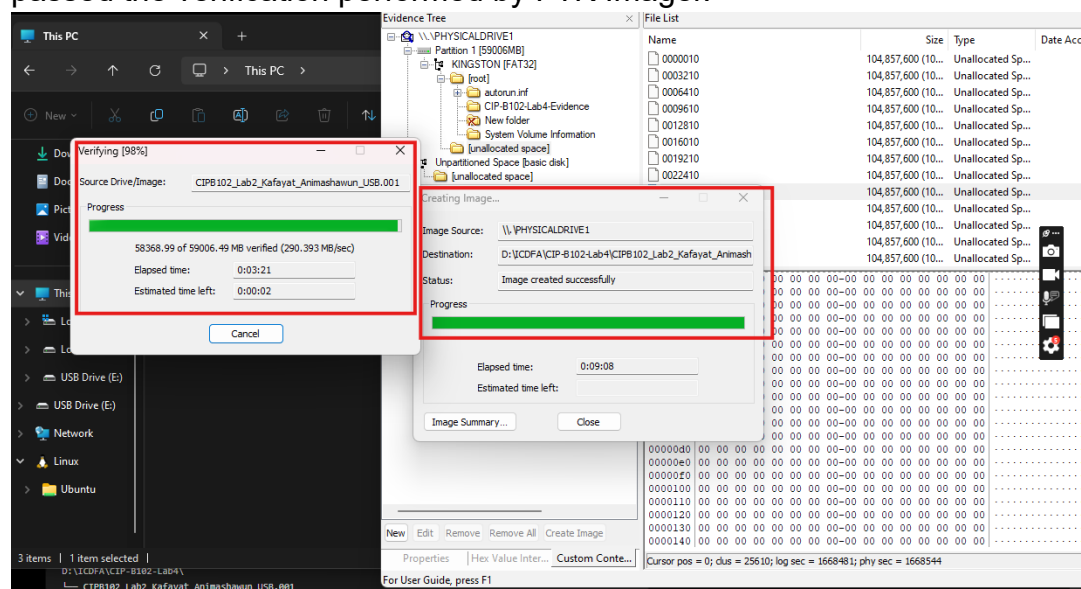


Figure 5: Successful completion of the forensic image acquisition and verification.

10. HASH VERIFICATION

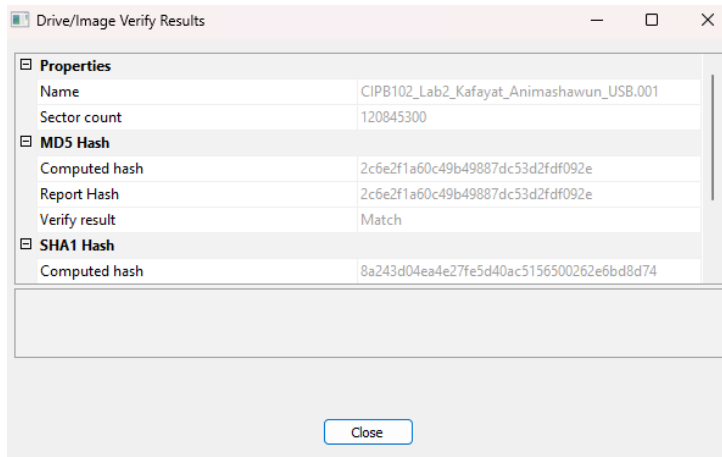
Cryptographic hashing was used to support the integrity verification of the forensic acquisition.

10.1 MD5

FTK Imager calculated the following MD5 hash for the source evidence:

2c6e2f1a60c49b49887dc53d2fdf092e

The verification report showed the same value as the report hash.



MD5 verification result: VERIFIED / MATCH

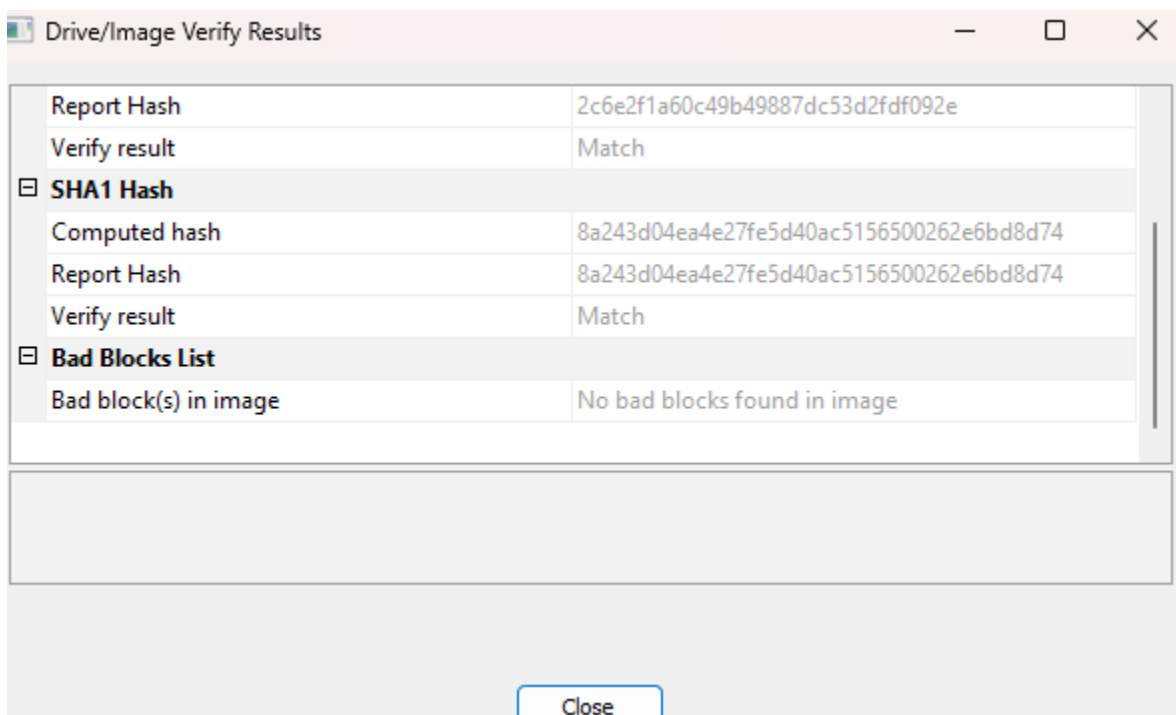
10.2 SHA-1

FTK Imager calculated the following SHA-1 hash:

8a243d04ea4e27fe5d40ac5156500262e6bd8d74

The verification report showed the same value.

SHA-1 verification result: VERIFIED / MATCH



10.3 SHA-256

The SHA-256 value calculated for the complete acquired image is:

SHA-256:

The SHA-256 result should be compared with the corresponding source-media SHA-256 value and documented in the final hash comparison table.

Hash Comparison Table

Hash Algorithm	Source USB	Result
MD5	2c6e2f1a60c49b49887dc53d2fdf092e	[MATCH]
SHA-1	8a243d04ea4e27fe5d40ac5156500262e6bd8d74	[MATCH]
SHA-256	14DF4D66849835BD29E20EF5502841AC2C79FC68CACE3E00B3B991592D7EA96B	[MATCH]

11. IMAGE VALIDATION

Following acquisition, the Raw/DD image was loaded into FTK Imager as an evidence item.

The image contained the expected partition and FAT32 file system. Within the file system, the folder:

CIP-B102-Lab4-Evidence was located.

The expected evidence file: kafayat_animashawun.txt was successfully located inside the acquired image.

FTK Imager also displayed the contents of the file, confirming that the evidence created before acquisition was preserved within the forensic image.

This validation demonstrated that the acquisition contained the expected evidence and was therefore suitable for further forensic examination.

Evidence File in Acquired Image

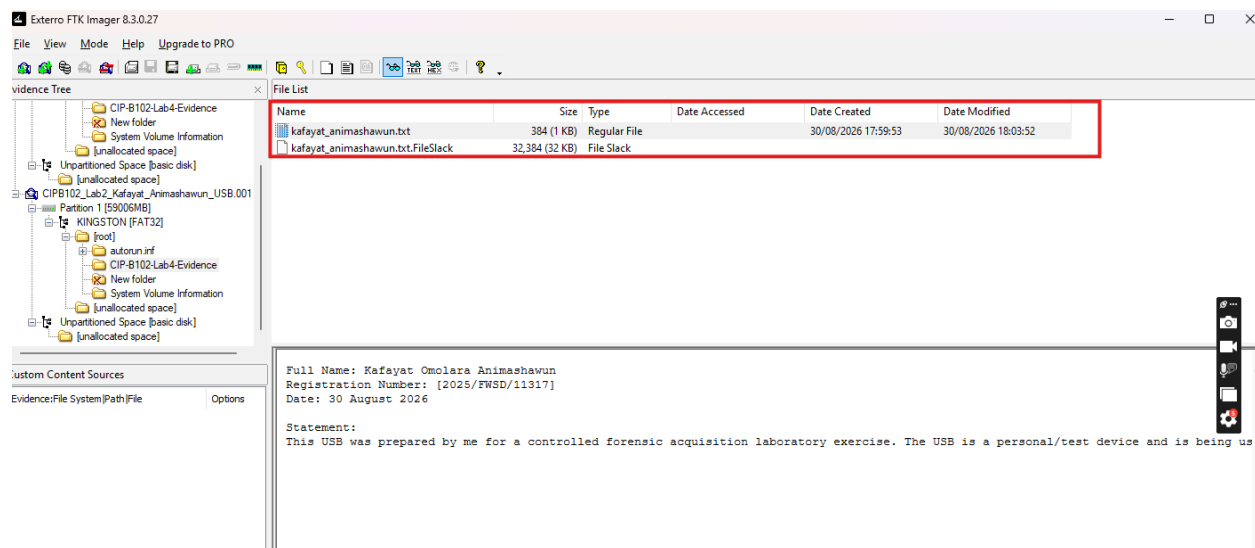


Figure 7: kafayat_animashawun.txt located within the acquired forensic image.

12. RISKS AND PRECAUTIONS

Several risks are associated with forensic disk imaging and low-level acquisition tools.

12.1 Selecting the Wrong Device

Selecting the wrong physical drive could result in acquiring the wrong evidence or, with certain tools and commands, overwriting important data.

Precaution: The USB was identified by its manufacturer, model, capacity and physical-drive identifier before acquisition.

12.2 Modifying the Source Evidence

Any modification to the original evidence could compromise forensic integrity.

Precaution: The source USB was used only as the acquisition source, while the forensic image was stored on the computer's D: drive.

12.3 Insufficient Storage Space

A forensic image of a physical device can be approximately the same size as the source device. Insufficient storage space can cause an acquisition to fail.

Precaution: Adequate storage space was made available on the destination drive before the successful acquisition was completed.

12.4 Interruption During Acquisition

Removing the USB or interrupting the imaging process could result in an incomplete or invalid image.

Precaution: The USB remained connected during the acquisition and verification process.

12.5 Image Integrity

An image could potentially become corrupted or altered after acquisition.

Precaution: FTK Imager's verification function was enabled, and the resulting MD5 and SHA-1 verification values were reported as matching.

12.6 Segmented Image Management

Raw/DD acquisitions may be divided into multiple segments. Missing, renamed or reordered segments could prevent the image from being correctly interpreted.

Precaution: All image segments generated by FTK Imager were retained in the same destination directory without renaming or modifying them.

13. REFLECTION

This practical provided hands-on experience with an important stage of digital forensic investigation: the preservation and acquisition of digital evidence.

One of the most important lessons from the exercise was the need to correctly identify the source device before performing forensic acquisition. A physical device identifier alone should not be relied upon without confirming its manufacturer, capacity and other available characteristics.

The exercise also demonstrated the importance of storing the forensic image separately from the original evidence. Using a dedicated destination folder on the computer helped prevent accidental modification of the source USB.

Another important lesson was the role of cryptographic hashes in demonstrating evidence integrity. The successful MD5 and SHA-1 verification results provided evidence that the acquired image corresponded with the source data processed during acquisition.

The practical also highlighted operational challenges associated with forensic imaging, particularly the need for sufficient storage capacity and uninterrupted acquisition. These considerations are important when performing forensic acquisition in a real investigation.

Finally, loading the acquired image and locating the expected evidence file demonstrated that successful acquisition is not sufficient on its own. The examiner should also validate that the resulting image can be accessed and that the expected evidence is present.

14. CONCLUSION

The USB image acquisition and hash verification practical was successfully completed using Exterro FTK Imager 8.3.0.27.

The Kingston DataTraveler 3.0 USB device was correctly identified as \\.\PHYSICALDRIVE1, with a source data size of approximately 59,006 MB. The device was acquired using the Raw/DD format and the resulting forensic image was stored separately from the source evidence on the local D: drive.

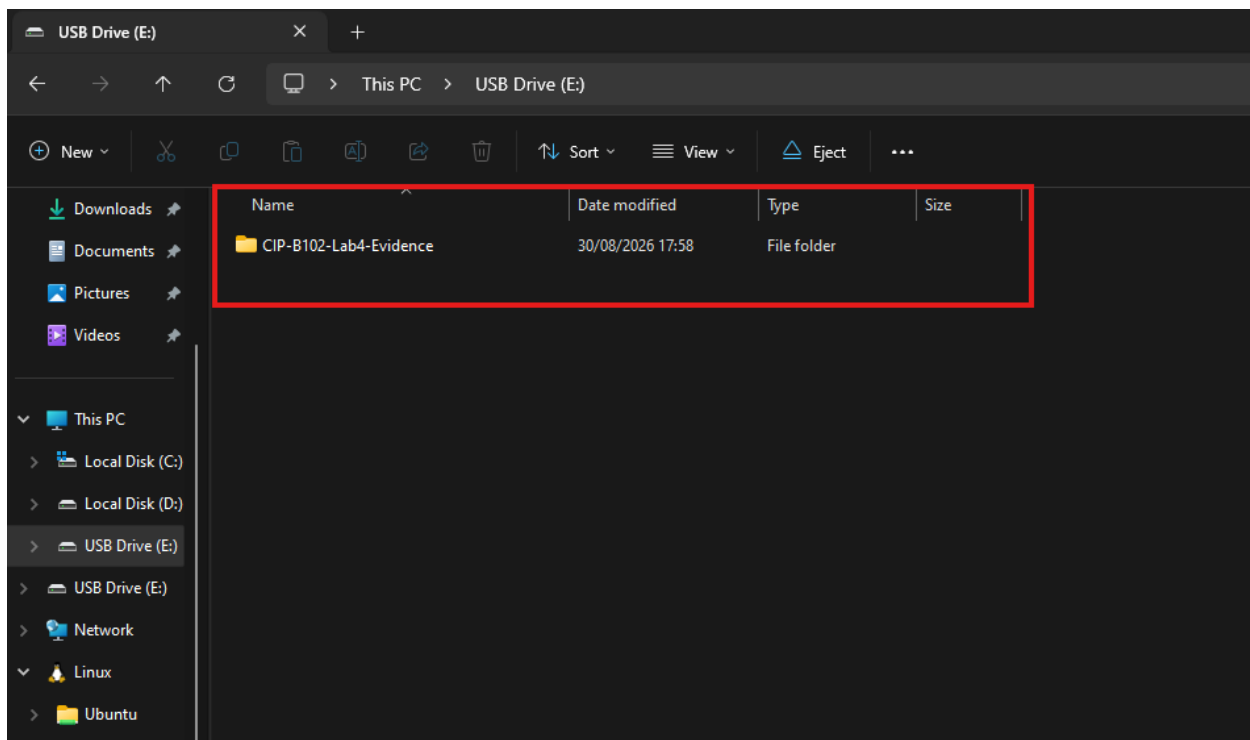
FTK Imager successfully completed the acquisition and verification process. The MD5 hash was 2c6e2f1a60c49b49887dc53d2fdf092e, and the SHA-1 hash was 8a243d04ea4e27fe5d40ac5156500262e6bd8d74. Both values were reported as verified matches. FTK Imager also reported that no bad blocks were found in the image.

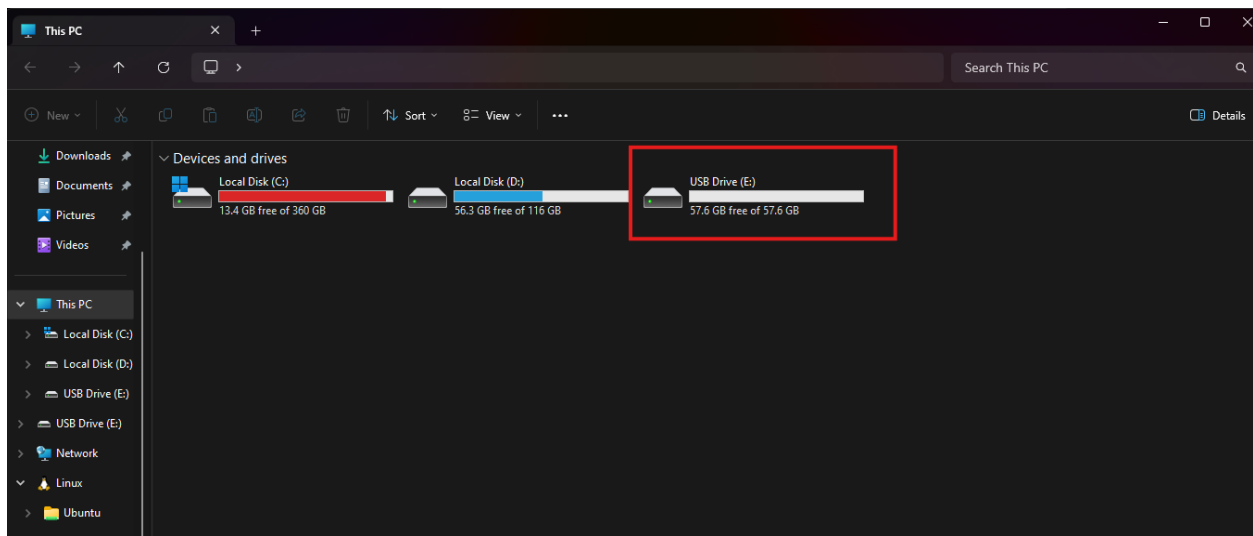
The acquired image was subsequently examined in FTK Imager, where the CIP-B102-Lab4-Evidence folder and kafayat_animashawun.txt file were successfully located. The contents of the evidence file were also accessible, demonstrating successful preservation and validation of the expected evidence.

Overall, the practical demonstrated the importance of careful source identification, controlled forensic acquisition, cryptographic integrity verification, image validation and professional documentation in digital forensic investigations.

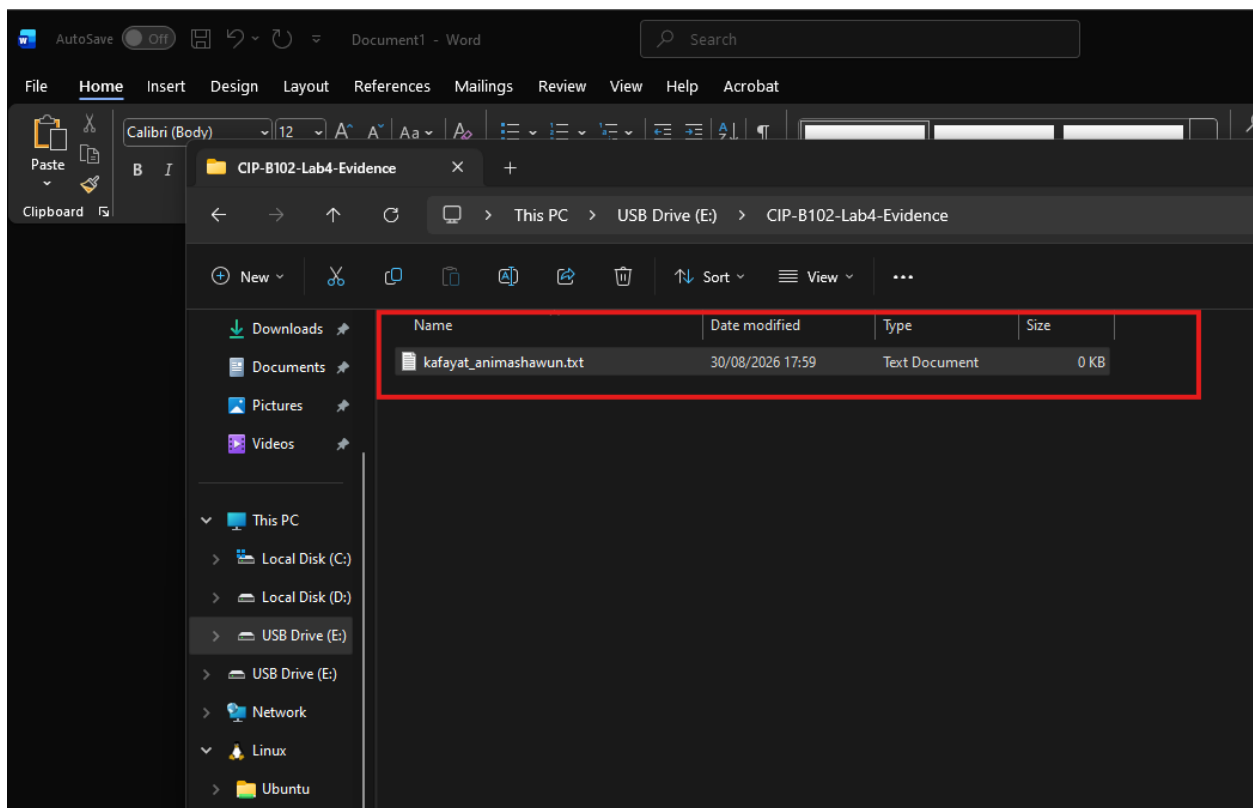
APPENDIX A: SCREENSHOT EVIDENCE

Flash drive evidence – Drive E

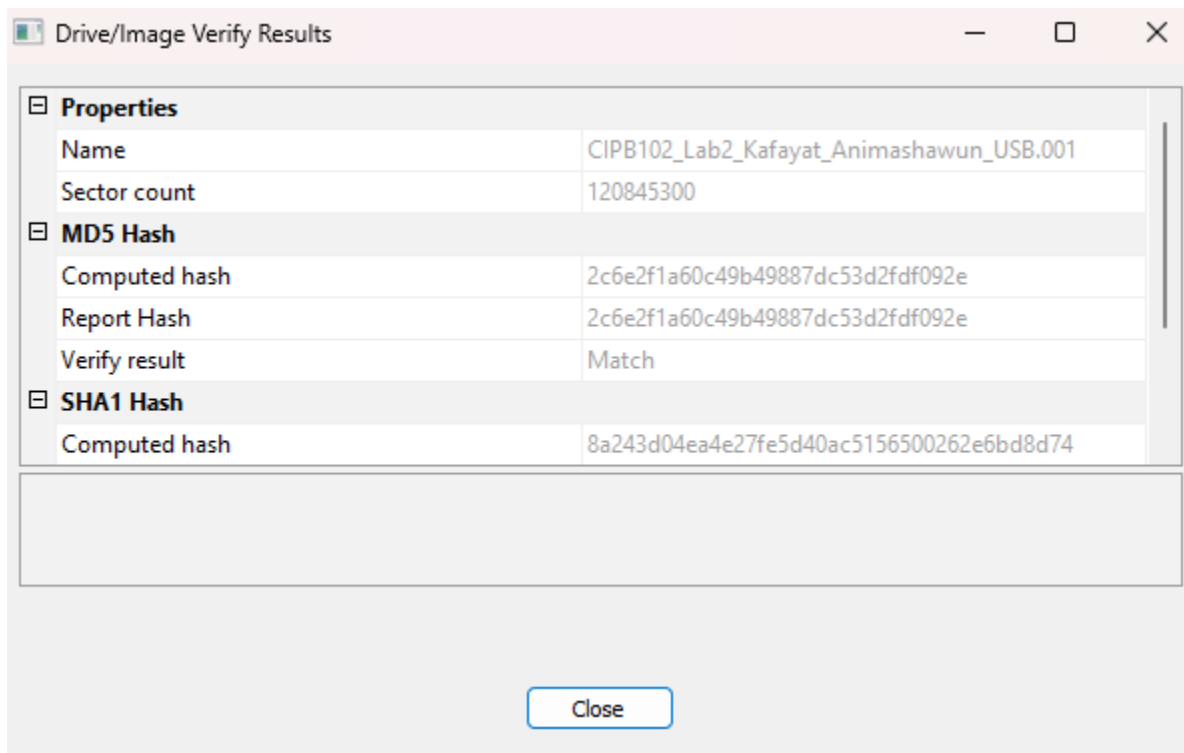
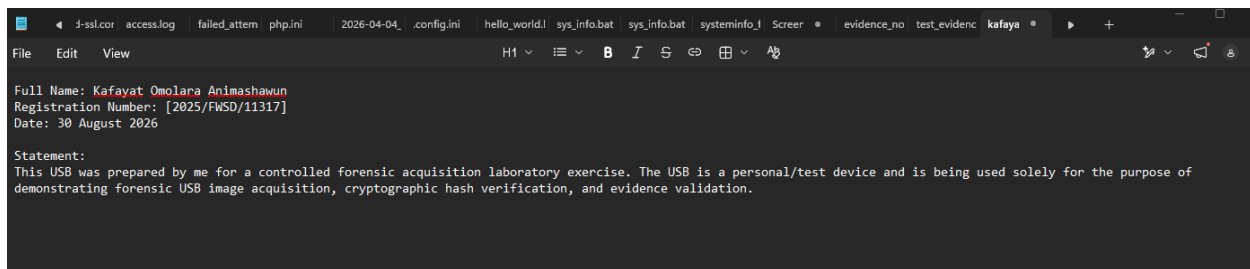




Evidence of text file creation

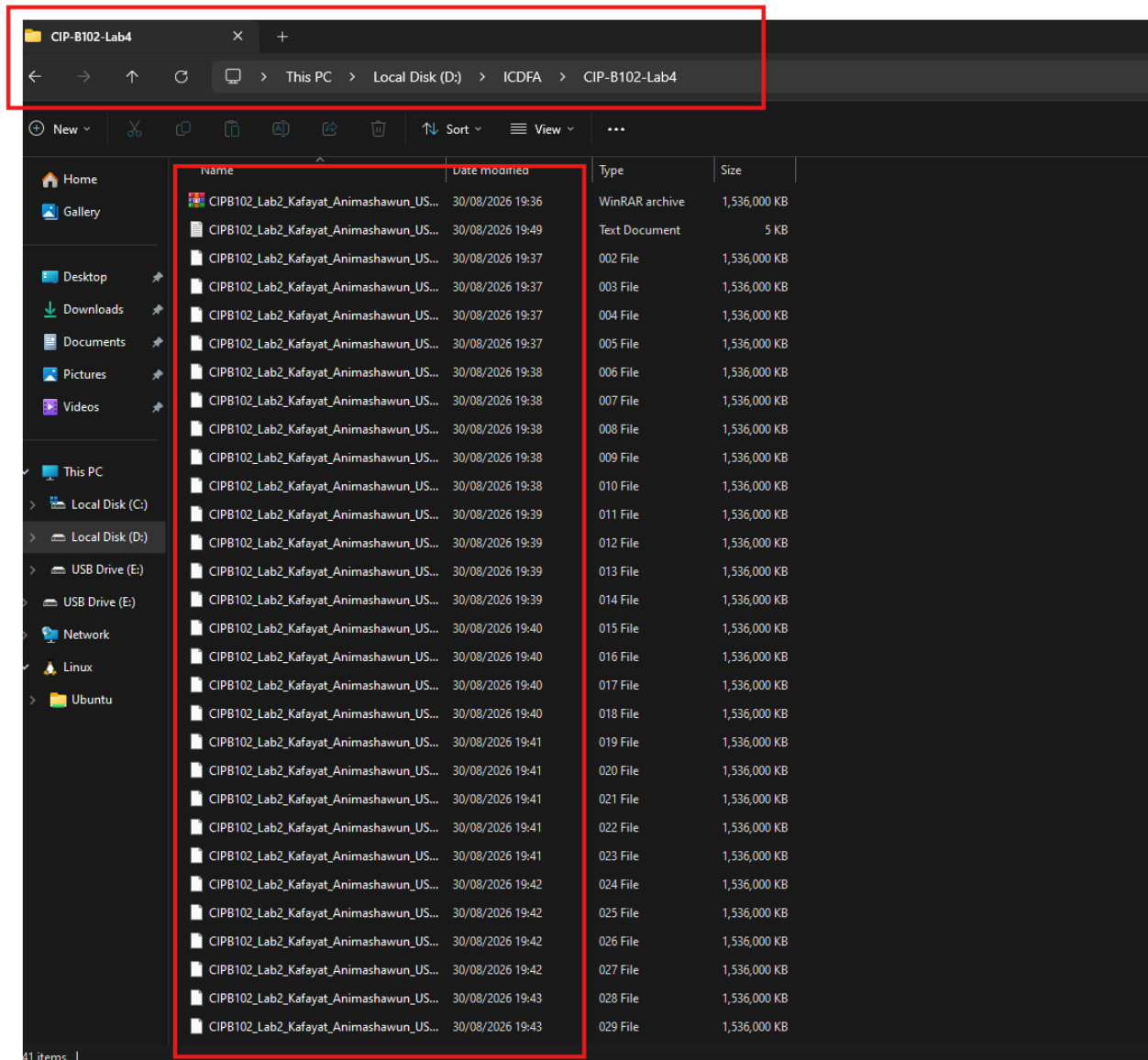


Text file content



SHA1 Hash	
Computed hash	8a243d04ea4e27fe5d40ac5156500262e6bd8d74
Report Hash	8a243d04ea4e27fe5d40ac5156500262e6bd8d74
Verify result	Match

MD5 Hash	
Computed hash	2c6e2f1a60c49b49887dc53d2fdf092e
Report Hash	2c6e2f1a60c49b49887dc53d2fdf092e
Verify result	Match



Algorit hm	Source USB	FTK verificati on
MD5	2c6e2f1a60c49b49887dc53d2fdf092e	Verified
SHA-1	8a243d04ea4e27fe5d40ac5156500262e6bd8d74	Verified
SHA- 256	14DF4D66849835BD29E20EF5502841AC2C79FC68CACE3E00B 3B991592D7EA96B	Verified

APPENDIX B: ACQUISITION INFORMATION

Tool: Exterro FTK Imager 8.3.0.27

Source: \\.\PHYSICALDRIVE1

Device: Kingston DataTraveler 3.0 USB Device

Source Data Size: 59,006 MB

Image Format: Raw/DD

Image Location: D:\ICDFA\CIP-B102-Lab4

Image Segments: .001 through .040

Acquisition Start: 30 August 2026, 19:36:38

Acquisition Finish: 30 August 2026, 19:45:46

Verification Start: 30 August 2026, 19:45:49

Verification Finish: 30 August 2026, 19:49:12

Bad Blocks: None reported

MD5: 2c6e2f1a60c49b49887dc53d2fdf092e

SHA-1: 8a243d04ea4e27fe5d40ac5156500262e6bd8d74

SHA256:

14DF4D66849835BD29E20EF5502841AC2C79FC68CACE3E00B3B991592D7EA96B